

DAVID (SEHEON) CHANG

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Education

University of Texas at Austin

Bachelor of Science in Computer Science

Bachelor of Science in Electrical and Computer Engineering

Aug 2023 – May 2028

Austin, Texas

University Honors

Experience

2nd Infantry Combined Division - U.S. 8th Army

Sergeant

Jun 2025 – Dec 2026

Pyeongtaek, South Korea

- Collaborated with the United Nations Command to provide tactical translation during ROK-US combined exercises. Also assisted by developing an off-line translator using cosine similarity, on ESP32, for military operations.
- Awarded the Army Commendation Medal and Army Achievement Medal by Major General Lombardo.

Ajou University School of Medicine

Research Intern

Jun 2024 – Sep 2024

Suwon, South Korea

- Translated R libraries into Python to hasten ETL of medical data into OHDSI CDM standards. Reduced processing time by 23% by preventing repeated merge and row-wise lookups: $O(N \cdot M)$ by utilizing hash join/maps: $O(N+M)$.
- Co-authored two clinical AI papers published in JMIR (JIF 6.0) utilizing BERT and GPT-based NLP pipelines.

LooxidLabs: CES 2022 Innovation Awards Winning Startup

Project Intern

Jun 2022 – Jan 2023

Seoul, South Korea

- Assisted develop a VR mental well-being service; showcased in NYC by gathering & monitoring upwards of 200 testers.

Projects

Ajou Course Manager | *Django, Docker, React, PDFMiner, OpenAI API, PostgreSQL*

Oct 2025 – Feb 2026

- Built & maintaining a course manager requested by MD professors. Currently being used at Ajou Univ. Medical School to manage 40+ professors, 200+ medical students, and 100+ courses with automated CRUD processes and analytics.

Tessera | *ESP32, C/C++, PN532 NFC, PySide6, AES-256-GCM, local SQLCipher*

Mar 2026 – Present

- Led a team to build an offline EHR dog tag system; placed 7th of 700 teams in the ROK Army Startup Competition.
- Engineered ESP32 firmware to read and decrypt compressed, AES-256-GCM encrypted records over SPI in under 1s, and built a PySide6/SQLCipher desktop application for offline serial provisioning.

Alpaca Trading Bot | *Python, SQLite (WAL Mode), Streamlit, systemd, Telegram API*

Jan 2025 – Present

- Engineered a 24/7 autonomous stock trading engine deployed on a Vultr VPS, managed via systemd unit files, utilizing SQLite Write-Ahead Logging (WAL) for telemetry tracking, a Streamlit dashboard, and Telegram alert webhooks.
- + 15% P&L fully automated without intervention, using 3 different strategies(RSI, Bollinger, short MA) over 2 months.

Sisypus | *C#, .NET 8, WPF, Windows Registry, Windows Internals (IFEO)*

Apr 2026 – Present

- Developed a focus utility that enforces app/website blocks, logs categorized activities, and surfaces AI-generated feedback. Integrated smart reminders (stretch, sleep, idle) and AI-assisted objective generation & to-do management.
- Reached 40+ users within one week of V1.0 release; maintained active user communication and shipped monthly updates driven by real-time feedback through <https://sisypusfocus.site>

Hearty | *Next.js, Python, AWS (Chalice - Lambda, API Gateway / CloudFront / S3), Neon Postgres*

Apr 2026 – Present

- Created a senior activity concierge using Gemini API. Built senior-friendly conversational AI navigation UI with Next.js single-page application hosted on S3 & CloudFront, backed by Python AWS Chalice microservices and Neon PostgreSQL.

Campus Involvement / Organizations

Faculty Applicant Interviewer

Chandra Family Department of Electrical and Computer Engineering

Spring 2024 – Present

University of Texas at Austin

- Recommended by faculty and advisory board. Processed 10 faculty hiring interviews for the ECE department.

Texas Convergent

IoT Build Team Tech Member

Spring 2024 – Present

University of Texas at Austin

- Built NFC alarm system only disabled when physically scanned. Pitched to +200 students and Austin-based VC.

Directed Reading Program

Natural Language Reasoning Team

Spring 2024 – Present

University of Texas at Austin

- Engaged in weekly research discussions on CoT and o1 reasoning, advised by UT CS Ph.D. student Zayne Sprague.

Additional Information

Languages: ARM Assembly, C, C++, C#, Java, JavaScript, Python, TypeScript | **Tools/Frameworks:** .NET, AWS, Django, Docker, Git, Kubernetes, Linux, Next.js, Node.js, PostgreSQL, React, SQLite, Streamlit